

PokeMart

A Commerce-Style Pokemon Search

Built with **Next.js (App Router)** + **Coveo Headless React (SSR)**

Repo: github.com/zikorao/CoveoProject

The Brief

Core requirements

- Facet search results by **Pokemon Type**
- Facet search results by **Pokemon Generation**
- Show each Pokemon's **picture** in the results

Stretch goals

- Server-side rendering, analytics, type-ahead, detail pages
- Make it look like a real commerce storefront

Solution at a Glance

- **Storefront UI:** search bar, type/generation facets, product-card grid, pagination
- **Clean data:** ~1025 Pokemon via Push API (`push API solution` source)
- **Type-ahead:** ML Query Suggestions + instant results (wildcard prefix)
- **Detail page:** `/pokemon/[name]` from Coveo Search API
- **ML:** QS + ART on `Search pipeline – pokemon-zikora` (hub `pokemon-zikora`)
- **Verification:** `test_art.py` + Relevance Inspector

Architecture

```
PokeAPI -> push_pokemon.py -> Push API -> Index  
Usage Analytics -> QS model + ART model  
Search pipeline - pokemon-zikora -> QS + ART  
Next.js: fetchStaticState -> hydrate -> SearchInterface  
SearchBox -> QS suggest + instant wildcard search  
scripts: simulate_searches | simulate_clicks | test_art
```

Tech Stack

- **Next.js 14 App Router, React 18, TypeScript**
- **@coveo/headless-react 2.9.16** (`/ssr` entrypoint)
- **Python 3** (stdlib only) for ingestion + analytics simulation
- **PokeAPI** as the upstream data source

Data Pipeline (Push, not Crawl)

```
scripts/push_pokemon.py
```

- One clean document per Pokemon: `title` , `type` (multi), `generation` , `picture`
- Push API **file-container** flow -> batch ingestion (~1025 docs)
- Credentials from env vars (no keys on disk); idempotent

Why push over crawl? Structured, reliable fields - no HTML noise in facets.

Coveo Configuration

- Engine reads `organizationId` + `accessToken` from env vars
- `analytics: { enabled: true }`
- **Constant query** isolates the clean source:
`cq = @source=="push API solution"`
- Controllers: `searchBox`, `resultList ( picture/type/generation )`,
`typeFacet`, `generationFacet`, `querySummary`, `pager`
- Live pipeline: **Search pipeline - pokemon-zikora** (search hub `pokemon-zikora`)
- QS + ART models associated on that pipeline

Search Experience

- Faceted filtering with **official Pokemon type colors** + counts
- Product-card grid: artwork, name, **generation badge**, **type chips**
- Query summary ("Showing 1-10 of N") and pagination
- Fully responsive storefront layout

Type-Ahead: Two Mechanisms

1. ML Query Suggestions (preloaded on focus)

- From the trained QS model; richer as traffic grows

2. Instant Results (live, as-you-type)

- Coveo matches whole words -> we use **wildcard prefix** (`char` -> `char*`)
- Debounced + cancellable (`AbortController`)
- Toggle via `INSTANT_RESULTS_ENABLED`

Together: works today (instant) + improves over time (ML).

Experiment: QS Cold-Start

1. QS model created -> status **Limited** (empty)
2. `querySuggest` API -> `completions: []`
3. `simulate_searches.py` -> **3,075** search events (`originLevel1: pokemon-zikora`)
4. Daily **rebuild** -> e.g. preload suggests `bulbasaur`

Lesson: QS needs analytics + rebuild; instant results bridge the gap.

Experiment: Instant Type-Ahead

- Coveo matches **whole words** -> `pik` returns 0
- Fix: `wildcards: true` + prefix query (`char*`)
- Debounced Search API calls in `SearchBox.tsx`
- Flag: `INSTANT_RESULTS_ENABLED` (on for production)

Experiment: ART Deployment

Issue	Fix
ART not in Relevance Inspector	Wrong pipeline / strict conditions
Token overrides searchHub	Match API key hub <code>pokemon-zikora</code>
Recommendation condition	Removed for catalog search
Search-only analytics	<code>simulate_clicks.py</code> (search + click)

Rebuild ART after aligned analytics.

ART Verified (**electric**)

- **66 results** - Kilowattrel, Pawmi, Bellibolt, ...
- Relevance Inspector: **Automatic Relevance Tuning** in query journey
- **Boost: 0** - ART ran, no extra promotion (simulated data / tied scores)
- **Pass criterion:** ART **executes**, not QRE > 0 in API

ML Toolkit (scripts)

Script	Purpose
<code>simulate_searches.py</code>	~3075 UA search events (QS)
<code>simulate_clicks.py</code>	~1025 search+click sessions (ART)
<code>test_art.py</code>	Checklist + searchUid for RI

`originLevel1` = `pokemon-zikora` (matches search hub)

Verify: test_art + Relevance Inspector

1. `python3 scripts/test_art.py` - checklist + **searchUid**
2. Admin -> **Relevance Inspector** -> paste UID
3. Confirm **Search pipeline - pokemon-zikora + ART** in query journey
4. **Boost: 0** is OK when ART executed (RI is authoritative)

SSR + Analytics

- **Hydration pattern:** `fetchStaticState` (server) -> `hydrateStaticState` (client)
 - Better first paint + SEO, no empty-result flash
- **NavigatorContext + middleware:** forwards referrer, UA, IP, stable visitor id
 - Analytics events correlate across SSR and client interactions

Security Hardening

- Hard-coded token -> `NEXT_PUBLIC_COVE0_*` **environment variables**
- `.env.local` gitignored; `.env.example` committed
- Exposed token **rotated** and **scrubbed from git history**
(`git filter-repo` + force-push)
- Push API key never stored in code

Challenges & Fixes (1/2)

Challenge	Fix
npm ERESOLVE (React 18 vs 19)	Isolated package.json, pinned React 18
UTF-8 build error	Rewrote files in ASCII
Coveo <code>fetch failed</code>	Corrected organization id
Missing navigator context	Added middleware + NavigatorContext
PokeAPI 403	Added User-Agent header

Challenges & Fixes (2/2)

Challenge	Fix
Polluted facet values (mixed sources)	Constant query to clean push source
QS empty (cold-start)	<code>simulate_searches.py</code> + rebuild
ART empty in RI	ART on pokemon-zikora pipeline; hub condition
Instant results 0 on partial input	Wildcards + prefix query
Analytics hub mismatch	<code>originLevel1: pokemon-zikora</code> in simulators
Secret in public repo	Env vars + rotation + history scrub

Key Trade-offs

- **Push API** over web crawl -> clean structured data
- **Constant query** over deleting the old source -> non-destructive, reversible
- **SSR + hydration** over pure CSR -> first paint + SEO
- **Instant results + ML QS** over QS-only -> beat the cold-start
- **Direct REST** on detail page over a second engine -> simpler

Roadmap

- `@type` "Include in results" -> inline type chips
- Richer QS prefix suggestions (more traffic)
- Stronger ART boosts (targeted clicks or real usage)
- ITD + richer document text for `lq` queries
- **RGA**: license, enable on pipeline, enrich content, UI (SSR upgrade)

Demo

- **Catalog:** facets, grid, pagination (`Search pipeline – pokemon-zikora`)
- **Type-ahead:** QS preload + instant (`char` -> Charizard, ...)
- **Detail page:** product card -> `/pokemon/[name]`
- **Verify ART:** search `electric` -> Relevance Inspector + searchUid

`npm run dev` -> <http://localhost:3000>

Thank You

PokeMart - Coveo Search Solution

github.com/zikorao/CoveoProject

Questions?